

## Volume 31

# Sentinel Decision Intelligence Engine (SDIE)

**Version:** 1.0

**Status:** Draft

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## Vision

Project Sentinel should not simply display findings.

It should help users prioritize, understand, and resolve them using evidence, business context, and organizational objectives.

The engine **does not make security decisions for the user**.

It provides transparent, explainable recommendations.

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## Philosophy

Every recommendation must answer five questions:

1. What happened?
  2. Why does it matter?
  3. What evidence supports this?
  4. What should be done?
  5. What happens if nothing is done?
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## Decision Object

Every recommendation becomes an object.

Decision ID

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## Decision Lifecycle

Created

Every recommendation can be tracked just like an incident or a mission.

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## **Recommendation Engine**

The engine combines:

- Evidence
- Trust
- Policies
- Asset Criticality
- Business Impact
- Mission Goals
- Historical Outcomes

to generate ranked recommendations.

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## **Explainability Engine**

Every recommendation includes:

### **Summary**

A plain-language explanation.

### **Technical Details**

Evidence and object relationships.

### **Impact**

Potential consequences.

### **Suggested Action**

Recommended remediation.

### **Verification**

How to confirm the issue is resolved.

Nothing is hidden behind an opaque AI score.

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## **Prioritization Model**

Instead of relying only on severity, Project Sentinel evaluates multiple dimensions:

## Action Plans

Recommendations can be grouped into executable plans.

Example:

### Improve Laptop Security

Tasks:

- Enable disk encryption.
- Update operating system.
- Remove unsupported software.
- Enable secure boot.
- Verify firewall configuration.

Progress is tracked automatically.

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## Organizational Recommendations

Examples:

- Complete MFA rollout.
- Replace unsupported operating systems.
- Review third-party access.
- Improve mobile enrollment.
- Strengthen backup verification.

The focus is organizational improvement rather than isolated technical fixes.

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## Personal Recommendations

For home users, recommendations become much simpler.

Instead of:

Registry key missing...

The platform says:

Your computer is healthy. Three improvements are recommended:

- Turn on automatic updates.
- Remove one application you no longer use.
- Enable device encryption.

Estimated time: **15 minutes**.

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## Decision History

Every recommendation records:

- When it was created.
- Who reviewed it.

- Which evidence supported it.
- When it was completed.
- Which mission verified it.

This builds organizational knowledge over time.

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## Decision Relationships

A recommendation can relate to:

- Assets.
- Missions.
- Policies.
- Incidents.
- Compliance controls.
- Evidence.
- Trust calculations.

Everything remains connected.

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## NEW CORE ENGINE

### Recommendation Learning Engine

The platform analyzes completed recommendations to answer questions like:

- Which recommendations are completed quickly?
- Which ones are repeatedly postponed?
- Which remediation steps consistently improve trust?
- Which recommendations produce little measurable improvement?

This helps organizations refine their operational practices over time.

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## NEW CORE ENGINE

### Decision Dependency Engine

Some actions depend on others.

Example:

Upgrade Operating System

Dependencies prevent unsafe or incomplete workflows.

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## NEW CORE ENGINE

## Outcome Verification Engine

A recommendation is not considered complete simply because someone clicked "Done."

The platform verifies success by collecting fresh evidence.

Example:

Recommendation:

Enable disk encryption.

Verification:

- Encryption status confirmed.
- Trust score updated.
- Mission completed.

Evidence closes the loop.

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## NEW IDEA

### Executive Decision Board

Instead of technical dashboards, executives see:

- Top organizational priorities.
- Expected security improvements.
- Compliance progress.
- Completed initiatives.
- Recommended next investments.

Every item links back to supporting evidence.

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## NEW IDEA

### Security Journey Planner

Organizations can define long-term objectives.

Example:

Year 1

- Asset inventory.
- Endpoint baselines.
- Mobile enrollment.

Year 2

- Zero Trust implementation.
- Cloud governance.
- Identity modernization.

Year 3

- Advanced analytics.
- Continuous assurance.
- AI-assisted investigations.

The platform measures progress against these goals.

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## NEW IDEA

### Decision Confidence Indicator

Every recommendation includes a confidence level based on:

- Quantity of evidence.
- Quality of evidence.
- Number of independent sources.
- Verification status.
- Consistency over time.

This helps users understand how strongly the available evidence supports the recommendation.

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## PS-ADR-0026

### Evidence Before Recommendation

**Status:** Accepted

#### Decision

Project Sentinel shall never generate a recommendation without traceable supporting evidence.

#### Rationale

Recommendations must be transparent, explainable, and auditable. Users should always be able to inspect the evidence and reasoning behind every suggested action.

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## One of the most important design principles

I want to formalize something that has guided our discussions from the beginning:

**Project Sentinel should reduce complexity, not increase it.**

Every new feature should make the platform easier to understand.

Not more complicated.

That means:

- New capabilities should fit naturally into the Mission model.
  - Every object should use the same consistent structure.
  - Every recommendation should be evidence-backed.
  - Every screen should help users answer: **What happened? What changed? What should I do next?**
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## Looking ahead

The next major blueprint should define the **Sentinel Workflow & Automation Engine**.

This will allow organizations—from home users to governments—to automate routine security work using visual workflows built on the same object, mission, and evidence model.

Combined with the Rule Engine and Decision Intelligence Engine, it will make Project Sentinel not just a system that observes security, but one that helps organizations continuously improve it in a controlled, transparent, and explainable way.

=====[ USER ]=====

Good and continue

=====[ ASSISTANT ]=====

Excellent. I think we are now designing one of the modules that will make Project Sentinel feel like a true operating system instead of a collection of security tools.

Until now we have built:

- Knowledge
- Missions
- Evidence
- Objects
- Digital Twin
- Trust
- Decision Intelligence

Now we need the component that connects everything together.

This becomes the **automation backbone** of Project Sentinel.

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## PROJECT SENTINEL BLUEPRINT

# **Volume 32**